

SUHARSHA CHEEDALLA

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M.S. Applied Data Intelligence, SJSU (May 2026) • AI Engineering • LLMs • Agentic Systems • RAG • Python

INTRODUCTION

AI engineer with an M.S. in Applied Data Intelligence (SJSU, May 2026) and 2 years of software engineering internship experience on a production financial platform. Built Atlas, an agentic customer support system combining a QLoRA fine-tuned SLM with RAG over 1.4M product records, evaluated at 72.3% task success across a 50-case G-Eval suite. Ships AI systems end to end: fine-tuning and evaluation, agent orchestration, FastAPI services, Docker, CI, and observability. Seeking full-time AI engineering roles.

EDUCATION

Master of Science in Applied Data Intelligence | San Jose State University May 2026
Coursework: Machine Learning, Generative AI & LLMs, Cloud Computing, Systems Integration, Advanced Data Analytics. Self-study: Designing ML Systems (Chip Huyen), Building Production RAG Systems (DeepLearning.AI), LangGraph & Multi-Agent Architectures (LangChain Academy).

Bachelor of Engineering in Computer Science | GITAM Institute of Technology May 2024
Coursework: System Design, Distributed Systems, Data Structures & Algorithms, Database Systems, Object-Oriented Programming, Operating Systems.

AI ENGINEERING PROJECTS

Atlas: AI Customer Support Agent *Python • FastAPI • Qdrant • PostgreSQL • QLoRA • Unsloth • HuggingFace • Docker*

- Built an agentic customer support system with deterministic routing across 8 domain tools, semantic retrieval over 1.4M product records in Qdrant, and PostgreSQL for transactional state, packaged with Docker Compose.
- Fine-tuned and benchmarked 4 small language models with QLoRA and Unsloth on real e-commerce support data, selecting Phi-4-mini-instruct on evaluation results rather than model size.
- Scored 72.3% task success and 3.79/5 G-Eval on a 50-case evaluation suite measuring relevance, faithfulness, completeness, and groundedness; the full harness, probes, and results are committed in the repo.

Tollgate: Cost-Aware LLM Router *Python • LangGraph • FastAPI • Groq • PostgreSQL • Alembic • Docker*

- Built a routing layer that scores query complexity with a deterministic, zero-cost classifier and dispatches each request to the right-sized model through a LangGraph supervisor, with per-request override support.
- Cut estimated inference cost by 81.6% against an all-premium baseline on a 40-query evaluation set, holding 97.5% tier-adjacent routing accuracy so no query lands more than one tier from its target.
- Shipped it as a real service: API-key auth, rate limiting, request validation middleware, Alembic migrations, per-request cost and latency tracking, and a pytest suite running in GitHub Actions CI.

LLM Jailbreak Detector *Python • scikit-learn • XGBoost • Streamlit*

- Trained a classifier that detects adversarial jailbreak prompts at 0.993 F1 on a 600-sample held-out test set, using 397 features including 13 handcrafted signals such as instruction-pattern counts, quote nesting depth, and imperative-verb scoring.
- Benchmarked logistic regression against Random Forest and XGBoost with cross-validation and hyperparameter search; committed ROC, precision-recall, and confusion-matrix artifacts back the reported numbers.
- Served the model through an API and a Streamlit dashboard for interactive adversarial testing.

PRODUCTION EXPERIENCE

Software Engineer Intern | Venhan Technologies (client: CHRIMS Inc.), Hyderabad, India Apr 2022 – Apr 2024

- Helped migrate a live Flask monolith to FastAPI microservices with zero downtime by rerouting endpoints incrementally through an Nginx reverse proxy using the strangler pattern.
- Kept API response times flat as the portal grew past 100,000 monthly transaction records by adding keyset pagination, composite B-tree indexes, and Redis caching for settlement queries.
- Cut live dashboard update latency 25% and held the 200ms SLA by replacing HTTP polling with WebSocket connections backed by Redis Pub/Sub.
- Brought multi-year settlement reports from minutes down to milliseconds by precomputing aggregations in PostgreSQL materialized views, refreshed by Celery jobs during off-peak hours.
- Reduced failed deployments by about 40% by building GitHub Actions pipelines that ran linting, pytest/Jest, and Docker builds to AWS ECR before every release.
- Stopped a recurring class of 500 errors from malformed track-hardware payloads by enforcing Pydantic validation at API boundaries, returning descriptive 400s instead of crashing workers.
- Cut time to detect production failures from hours to under 3 minutes by instrumenting services with Sentry and Grafana dashboards and setting alerts on latency, error rates, and connection pool usage.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Bash

AI/ML: LangGraph, LangChain, RAG, QLoRA/LoRA Fine-Tuning, HuggingFace, LLM Evaluation (G-Eval, Ragas), scikit-learn, XGBoost, PyTorch, Prompt Engineering

Software Engineering: FastAPI, React, REST APIs, WebSockets, Docker, Pydantic, Pytest, CI/CD, Git/GitHub Actions

Cloud & Data: AWS (ECS, ECR, S3), PostgreSQL, Qdrant, Pinecone, Redis, Kafka, Prometheus, Grafana

Certifications: AWS Cloud Foundations • Claude 101 (Anthropic) • AI Fluency (Anthropic) • B.E.L.L.A x Mule Run Hackathon Winner